Rise in

Pre-Workshop Setup Guide

Stellar

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Introduction

This guide is created to help participants set up their development environment before attending the workshop. To ensure the workshop runs smoothly, participants are expected to have installed several tools that will be used during the session.

Setup Guide for Windows

Smart contracts on Stellar are written using the Rust programming language.

Before starting to build smart contracts on Stellar, there are several tools that need to be installed first:

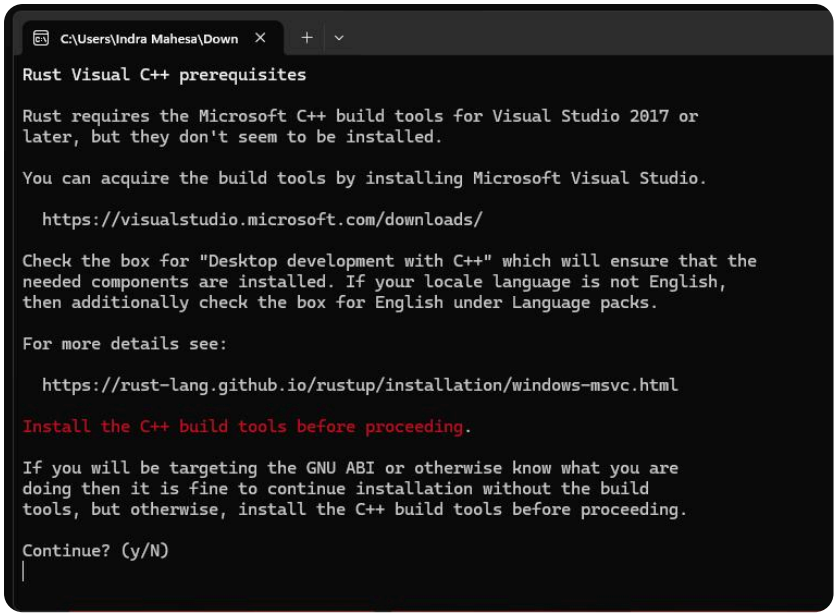
- Rust Toolchain
- Microsoft C++ Build Tools
- Stellar CLI

If you are already using WSL (Windows Subsystem for Linux), you can also follow the installation guide for Mac/Linux.

- **Install rustup**

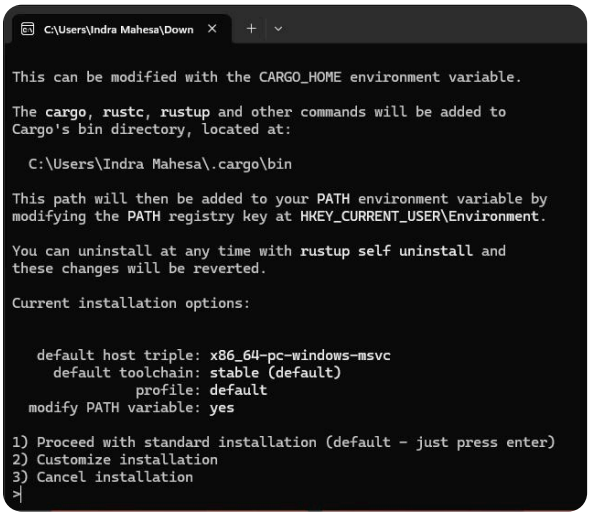
Download **rustup** from the link below:

https://rustup.rs/



After running the .exe file, in the terminal (as shown above), type **y** and press **Enter** to continue the installation process.

Setup Guide for Windows



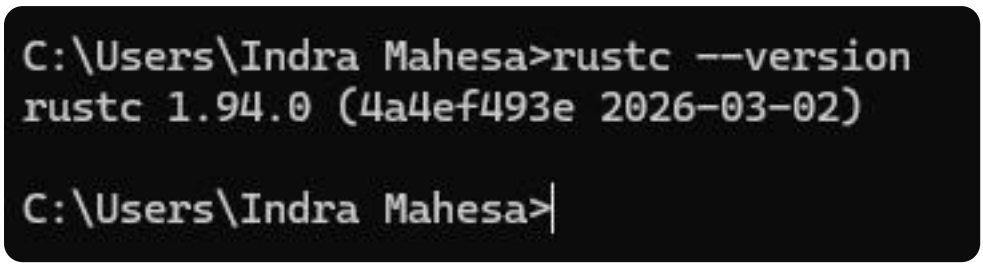
Next, you will see installation options. Simply press **Enter** to proceed with the default settings.

- **Verify the rustup installation**

Open **Command Prompt (CMD)**, then type the following command:

rustc --version

If a version number appears and there are no errors as shown in the image below, it means Rustup has been successfully installed.



Smart contracts on Stellar require rustc version 1.84.0 or newer.

- **Install rust target**

Open **Command Prompt (CMD)**, then type the following command:

rustup target add wasm32v1-none

Setup Guide for Windows

If successful, the result will appear as shown in the image below.

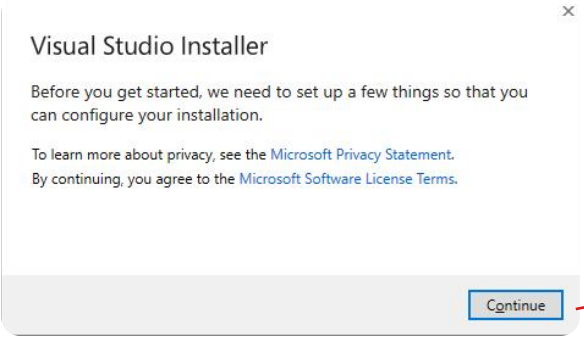
```
C:\Users\Indra Mahesa>rustup target add wasm32v1-none
info: downloading component rust-std
     rust-std installed                                11.15 MiB
C:\Users\Indra Mahesa>
```

- **Install Microsoft C++ Build Tools**

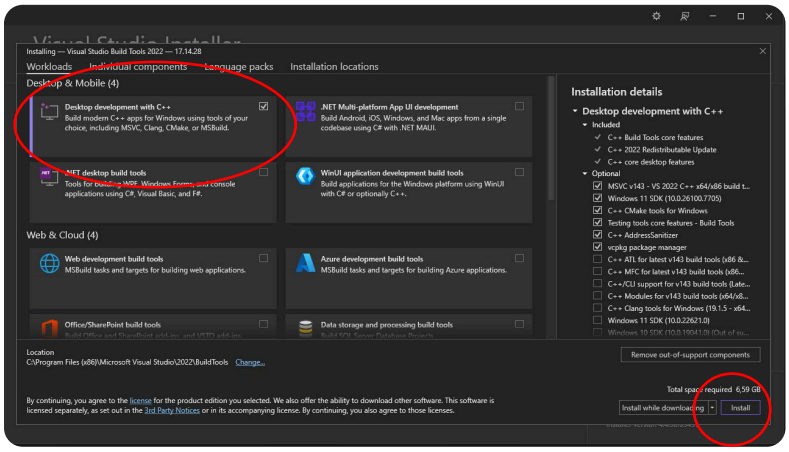
Stellar CLI requires Microsoft C++ Build Tools to run on Windows.

Open **Command Prompt (CMD)**, then type the following command:
curl -L -o vs_buildtools.exe https://aka.ms/vs/17/release/vs_BuildTools.exe

After the download is complete, run the following command:
.\vs_buildtools.exe



When the program runs, a GUI will appear as shown in the image above. Click Continue



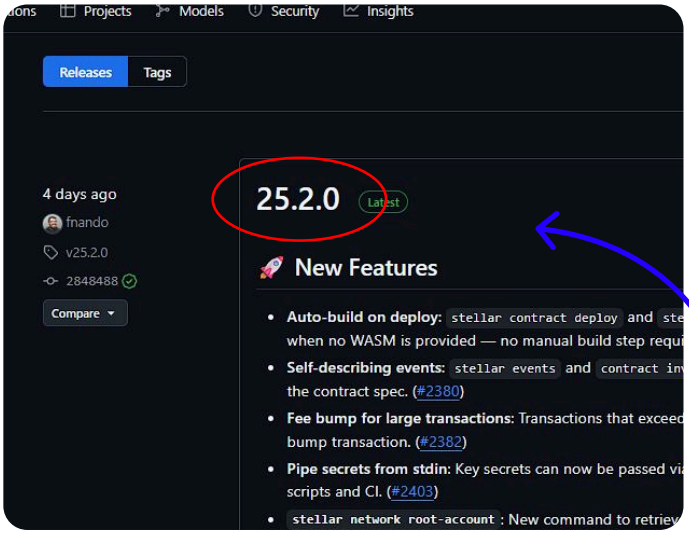
Check the **Desktop development with C++ option**, then click Install to continue the installation process.

Setup Guide for Windows

- **Install Stellar CLI**

Check the latest version of Stellar CLI at the link below:

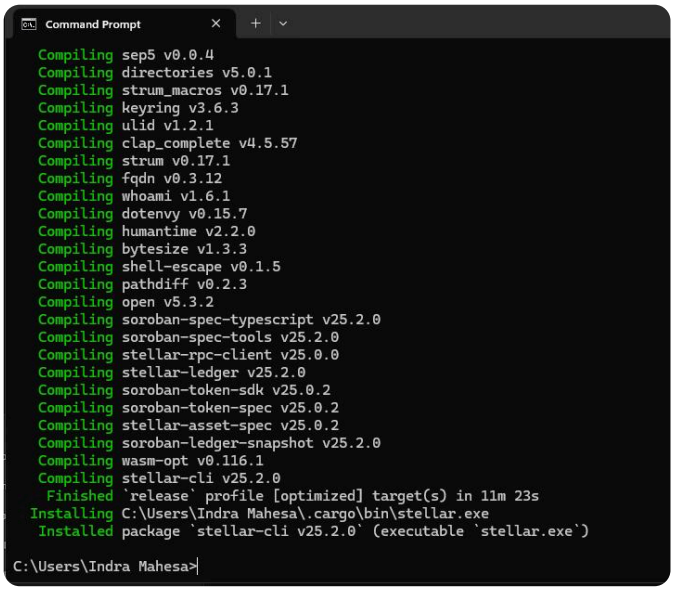
https://github.com/stellar/stellar-cli/releases



If the latest version is different, please adjust the version number in the CMD command below accordingly.

Open **Command Prompt (CMD)**, then type the following command:

cargo install --locked stellar-cli@25.2.0



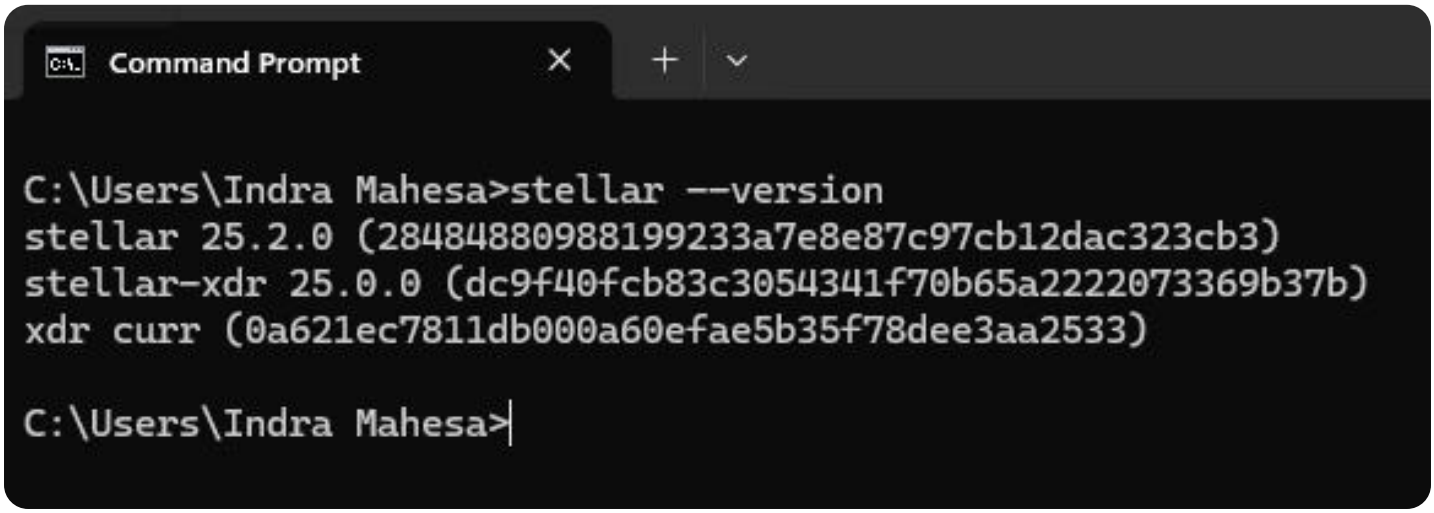
If the terminal display matches the image shown beside it, then the Stellar CLI download process has been completed.

Setup Guide for Windows

- **Verify Stellar CLI Installation**

Open **Command Prompt (CMD)**, then type the following command:

stellar --version



If your terminal output matches the image above and there are no errors, then Stellar CLI has been successfully installed on your computer!

If you encounter any issues during the setup process, you can contact the admin through the Rise In SEA Telegram group.

Setup Guide for Mac/Linux

Smart contracts on Stellar are written using the Rust programming language.

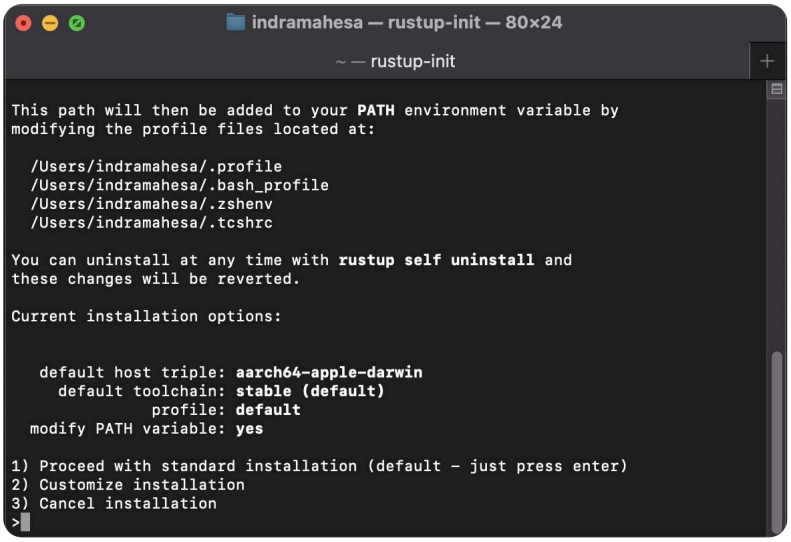
Before starting to build smart contracts on Stellar, there are several tools that need to be installed first:

- Rust Toolchain
- Stellar CLI

- **Install rustup**

Open **Terminal**, then type the following command:

curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh



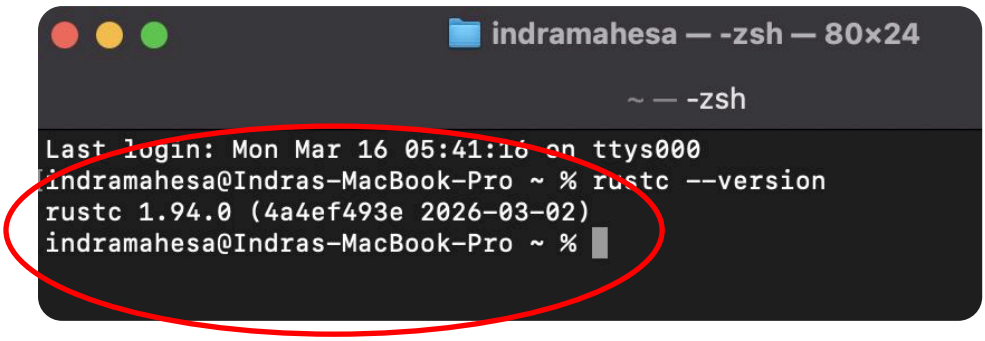
Next, you will see installation options. Simply press **Enter** to proceed with the default settings.

- **Verify rustup installation**

Close the terminal and reopen it, then type the following command:

curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh

Setup Guide for Mac/Linux

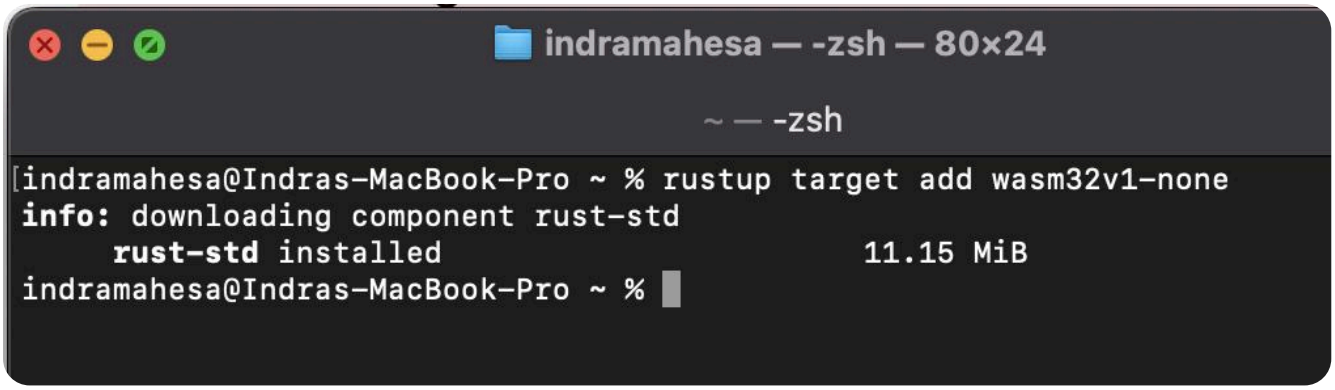


If a version number appears and there are no errors as shown in the image above, it means Rustup has been successfully installed.

- **Install rust target**

Open **Terminal**, then type the following command:

rustup target add wasm32v1-none



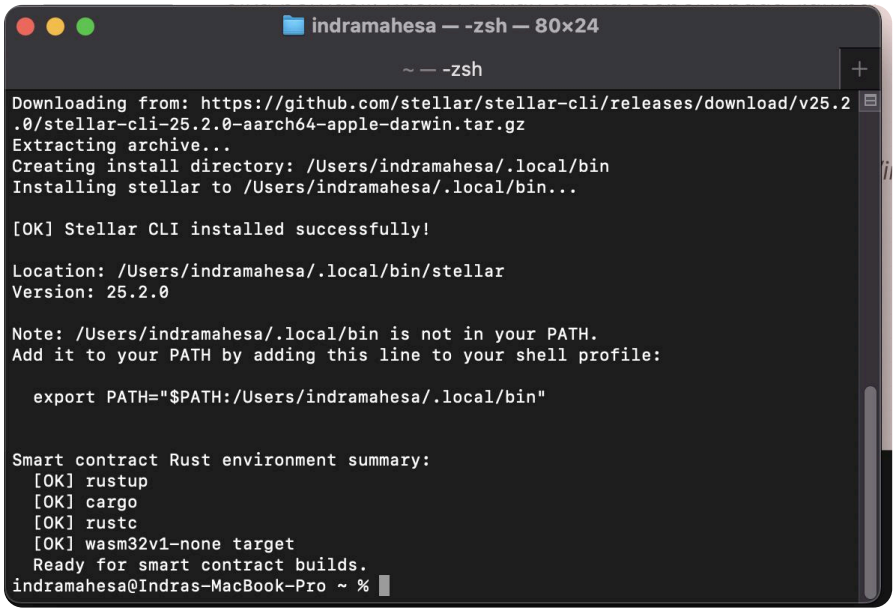
If successful, the result will appear as shown in the image above.

- **Install Stellar CLI**

Open **Terminal**, then type the following command:

curl -fsSL https://github.com/stellar/stellar-cli/raw/main/install.sh | sh

Setup Guide for Mac/Linux



If the output matches what is shown above, the next step is to add the Stellar CLI path to the environment variable.

Run the following command in **Terminal**:

1. *echo 'export PATH="$PATH:$HOME/.local/bin"' >> ~/.zshrc*
2. *source ~/.zshrc*

For Debian/Ubuntu Linux users, you must run the following command in the **terminal**:

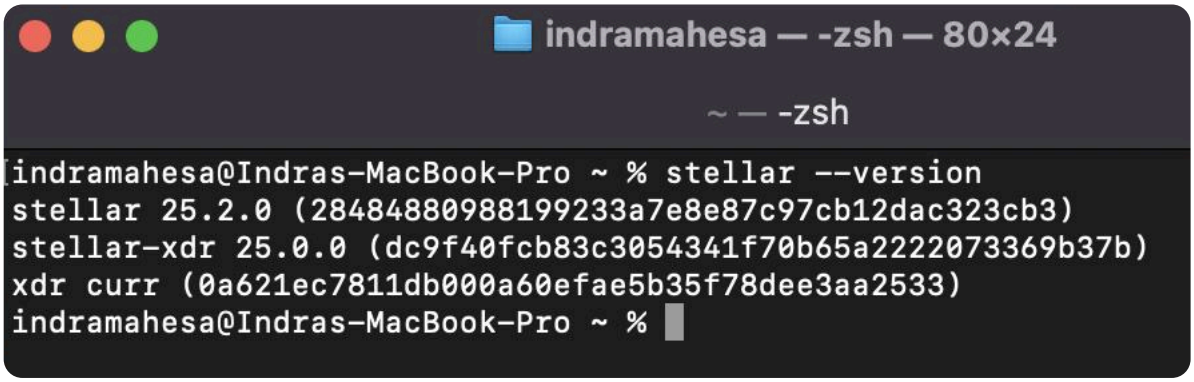
sudo apt update && sudo apt install -y build-essential

Setup Guide for Mac/Linux

- **Verify Stellar CLI Installation**

Open **Terminal**, then type the following command:

stellar --version



If your terminal output matches the image above and there are no errors, then Stellar CLI has been successfully installed on your computer!

If you encounter any issues during the setup process, you can contact the admin through the Rise In SEA Telegram group.

References

Set Up and Configure Your Environment for Writing Smart Contracts on Stellar

https://developers.stellar.org/docs/build/smart-contracts/getting-started/setup

Create a simple Smart Contract “Hello World”

https://developers.stellar.org/docs/build/smart-contracts/getting-started/hello-world